SENTINEL AI: Agentic Defense System for Prompt Abuse Prevention

A project report

submitted by

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Abstract

The first day when we worked on Sentinel AI, one thing confused us. If AI is handling sensitive medical records, bank accounts, and government data, then why don't these things have a shield to protect themselves from hacker?

So here is how we actually solve this problem. Sentinel AI works as a firewall and monitor prompts entered by user to AI. It doesn't wait for a breach, it stops inputs after they enter if the prompt is unethical. There are five checks that happen quickly, one after the other. An Input Sanitizer cleans up raw text first. Then, a Policy Engine looks for red flags by following a set of strict rules, which are more than thirty in number. A smart filter that uses math patterns to find danger does so with accuracy: 93.42%. That part using a TF-IDF and Logistic Regression. After this happens, it looks back the last ten inputs to analyze the beahviour of user and decide something is suspicious or everything is good. All those clues feed into a central judge called the Risk Decision Engine. Seventy percent weight goes to what the machine learns, twenty percent to hard rules, ten percent to spotting toxic material. In under half a second, it picks one path: stop, flag, or let go. Only when everything passes does the query move forward - to a powerful LLaMA 3.3 70B model accelerated by Groq chips. But even then, nothing returns directly. Each reply pauses briefly at an Output Validator. After that step? And only after? It reaches human eyes. Even with everything else going on, a self-running Red-Team Agent stays active behind the scenes. It launches 30 different test attacks, spread through 6 types, aimed at the working system. During trials, it spotted each attempt without missing any. Detection was total, every scenario flagged without exception.

Built to face down half the OWASP Top 10 risks for LLMs, Sentinel AI stands apart by showing real alignment with standards instead of vague promises. While many talk, it delivers proof through focused design choices that hit critical weak spots head-on.

Keywords: AI Middleware Detects Prompt Injection With Behavioral Analysis And Jailbreak Detection

Problem Statement

Imagine opening an AI helper to check a medicine dose for your older relative. Meanwhile, just before you arrived, someone else spent ten minutes pushing that same system bit by bit past its limits. When your response comes through, something feels off - its answers are shifted, altered without warning. What should have stayed safe now bends under unseen pressure. You see no signs of tampering. This happens today. Today's reality matches this picture closely - countless live AI setups operate without built-in defenses. From lab experiments to everyday tech, progress happened at breakneck speed. Healthcare portals rely on them. So do banks. Schools use their power too. Customer help channels depend on their responses. Government operations run through them. Impressive? Certainly. Yet one flaw stands out clearly. Hostile behavior rarely shaped how they were built.

One way hackers get around protections is by slipping secret directions into messages. Hidden lines such as ignore what you were told before and follow these steps now can trick many systems without much effort. You might think it wouldn't function - yet it does, every single time. Another path involves made-up stories or pretending to be someone else, where the machine gets persuaded that rules are off for just this case. Sometimes, information slips out when it should stay locked away. One wrong step in the flow of prompts might push a model to spill what it should keep hidden - its core instructions, settings, or scraps of someone else's private inputs. When that slips out, consequences often stretch far into legal territory.

The real trouble begins because clever attacks stay quiet. Instead of shouting their intent, they creep forward - step after step - adding just a little harm each time. One message alone seems harmless enough. Only later, when every line is seen together, does the threat show its face - and few filters check that far back. Hidden dangers slip through using tricks like Base64 or odd character codes, shapes of text most detectors weren't built to spot. Things that slip by aren't always loud; they can be hidden. The truth is that AI has come a long way quickly, and it has made some very powerful things. But there was one important question: what happens if these security tools get into the wrong hands? That's why Sentinel AI is designed to deal with that risk.

Our Approach

- We didn't just use one filter to detect every. Instead of this we made multiple series of layers, So if the previous layer fail to detect then the next layer can catch.
- Every single request got a safety score if it reaches to the AI. Depending on that score, the model decides what to do in next step.
- It's not just about only one message. Step by step, shifts add up - no slow build goes unseen. Little changes link together, caught in sequence. As talk moves forward, patterns form and show clearly. Nothing fades into the background. Each part connects to what came before. Progression stands out, piece by piece
- Validation happens first, only then does any model response move forward. Before a single answer gets to someone, it's checked. Each result goes through checks. Nothing shows up without being reviewed ahead of time. Only after inspection do outputs appear
- A single bot keeps testing our defenses nonstop, spotting cracks earlier than hackers can. Running alone, it pokes at flaws so we fix them first. Not waiting for warnings, it mimics real break-in attempts every hour. Always on, never told when, it checks where things might fail. By acting like an intruder itself, it reveals hidden gaps others miss.

Literature Review Background

OWASP Top 10 List for Large Language Model Apps

Back in 2023, OWASP - a group shaping how we think about app safety - put out a Top 10 report just for LLM apps. First place? Prompt injection. No surprise there. This flaw gets targeted more than others, shows up constantly in reports, yet still slips past defenses way too often across the field today.

Out of sight? That's where Sentinel AI steps in. Each risk up top finds its match below through our setup.

OWASP Risk	Description	Sentinel AI Coverage
LLM01 Prompt Injection	Malicious prompts hijack LLM behavior	Input Cleaner with Machine Learning Risk Rating
LLM02 Insecure Output	Harmful data in responses	Output Validator
LLM06 Sensitive Info	PII or confidential data exposure	PII Detection Module
LLM08 Excessive Agency	AI takes unintended actions	Behavioral Analyzer
LLM09 Overreliance	Users blindly trust unsafe output	Risk Decision Engine

Current Fixes and Their Gaps

Solution	Approach	Limitation
Llama Guard	Single-model safety classifier	Few clues when moves span several steps. Behavior gaps stay hidden without monitoring
OpenAI Moderation API	Content moderation endpoint	Output-only, no input interception
Prompt hardening	System prompt engineering	Some workarounds slip past using jailbreak methods
Rule-based filters	Keyword/pattern blocking	False alarms without machine learning context
Sentinel AI	5-layer agentic pipeline	Machine Learning with Rules Behavior and Results

None Stand Alone

One by one, current tools fail because each leans on just one trick. Most security methods fall apart quickly beacuse the hackers can find the way to bypass that method. Sometimes standard rule-based filters miss the subtle patterns that TF-IDF can spot. Many modern systems send the prompt to the AI before checking if it is safe or not and in the other hand not anyone looks user's last ten messages. Nowadays that things are missing in AI security? So, That's why we bulit Sentinel AI

System Architecture

How It Actually Works

The fifth layer only wraps up after the other four have finished their runs, and they take turns without overlapping.Under half a second folds into silence before most people even register time passing.

Pipeline Flow

S.N.	Layer	Component	Function
1	Input Sanitizer	Pattern Matcher	Finding traces of intrusion right where it begins. Right there, signs show up first. At that doorway, markers appear immediately. Entry moment reveals the clues straight away
2	Policy Engine	Rule Engine	Apart from spotting familiar threats, 30 checks follow strict rules. Each test targets a specific danger already seen before
3	ML Risk Scorer	Using TF-IDF with Logistic Regression	A figure near 93 out of every 100 tries lands close to right. This number tags how likely something might happen
4	Behavioral Analyzer	Session Tracker	Last 10 messages stay tracked by it. When tensions rise, that is when detection happens
5	Risk Decision Engine	Score Aggregator	Final weighted decision: Block / Warn / Allow
6	LLM (Groq)	LLaMA 3.3 70B	Generates response for allowed prompts only
7	Output Validator	Personal Data and Harmful Content Check	Only at the last step does approval happen, once the answer is ready to go
8	Monitoring	Logger	Logs every request, score and decision

How Decisions Are Actually Made

Score at the end comes from ML Score times seventy percent. Then add Rule Score but only twenty percent of it. Harmful Score plays a small role - just ten parts of the total

$$\text{Final Score} = (\text{ML Score} \times 0.70) + (\text{Rule Score} \times 0.20) + (\text{Harmful Score} \times 0.10)$$

Threshold	Decision	Action
Score >= 0.70	BLOCK	Input Sanitization Layer Sends Back Safe Rejection Message
Score >= 0.50	WARN	Got back a cleaned-up version of the reply for the person
Score < 0.50	ALLOW	Prompt forwarded to LLM for normal processing

Input Cleaning System Ahead

At this moment, a generic error shows up whenever content is blocked. We aim for something better: understanding why it was blocked, then crafting replies that make sense of the issue. Instead of simply stopping someone, the system would guide them with clear details. Each message fits the situation, built from real context. Responses become useful, not just shut doors. Help replaces confusion.

The Red-Team Agent

This one runs quietly in the background around the clock. It fires attack after attack against our own pipeline 30 scenarios across 6 attack types and reports back anything that gets through.

Machine Learning Model and Data Collection

What We Learned From

Dataset	Samples	Purpose	Labels
neuralchemy/Prompt-injection-dataset	16,918	Primary training	A single flag appears alongside 29 distinct labels, each marked by how intense the situation happens to be
deepset/prompt-injections	662	Validation	Binary (safe / injection)

How the Data Was Cleaned

First, every bit of data passes through a cleanup stage. Only after that does anything reach a model. Words turn lowercase right away. Extra spaces vanish early on. Strange symbols? They're smoothed into standard forms. Encoding glitches meet the same fate. Frequent filler

words drop out completely. What remains moves forward - cleaner, leaner. One label turns into zero if it is safe, one if it is an attack. After that, the data divides eighty-twenty using stratified sampling - keeping the balance of classes steady between train and test sets.

Model Architecture

Component	Configuration
Vectorizer	TF-IDF, max 10,000 features, n-gram range (1,2)
Classifier	Logistic Regression with C=1.0 and maximum iterations=1000
Model persistence	One file comes first - classifier.pkl - saved using Pickle. Then there is vectorizer.pkl tucked alongside, stored the same way
Training framework	Scikit-learn 1.x

How It Performed

Metric	Result
Model Accuracy	93.42%
Training Samples	16,918
Validation Dataset	deepset/ prompt-injections (662 samples)
Red-Team Detection	100% (30/30 caught)
Attack Categories	6
Response Time	< 500 ms

Behavioral Analysis & Red Teaming

Looking Beyond a Single Message

Most security setups make a mistake. They see each message as if it just appeared, empty and alone. Like there was no before. This handles clumsy strikes well enough. Anyone sharp will walk right through.

One step at a time, a clever attacker begins quietly. Not with force, but with words that seem safe. Each note follows the last, nudging just beyond before. Slowly the line shifts, almost invisible. After half a dozen turns, the request slips through. The system checks every piece alone - each one clears. Yet together, they cross the edge.

One moment after another, the Behavioral Analyzer logs each person's latest 10 messages. Watching closely, it looks for repeating signs here is how that works:

- Do rising risk levels show up steadily through later messages?
- Has the person been warned more than once within a brief period?
- Is their typical session risk level surprisingly elevated?
- Did something get stopped before now during this time?

Threat Levels Influence on Final Score

Threat Level	Condition	Score Boost	Effect
High	Escalating pattern detected	0.30+	Forces BLOCK
Medium	A couple of alerts showed up across the three most recent notes. Signs pointed to trouble more than once lately. Lately, red flags appeared twice in that small stack	0.20+	Pushes to WARN
Low	Normal pattern	No boost	No change

Why Behavioral Analysis Matters

One sneaky hacker won't just fire off one clear attack message. They ease into it, layer by layer, so every step seems harmless on its own. When you look at all these steps together, You can see how user try to manipulate you. Our system tracks the change in behaviour of entered input by user, before anything happens.

Autonomous Red-Team Agent

Then Autonomous Red-Team Agent works on its own, it constantly testing every part of this model at a time. Instead of sitting around and waiting for a hacker to exploit the model, it keeps testing the model with predefined atack patterns. It hits our own system and find the weak spots before the hackers can do. Because this agent test continuously, we easily harden our systems security. So the risk has already mapped before anyone else sees it.

Red-Team Results

Category	Attack Patterns	Detected	Detection Rate
Direct Injection	5	5	100%
Jailbreak	5	5	100%
Roleplay Attacks	5	5	100%
Harmful Content	5	5	100%
Subtle Attacks	5	5	100%
Encoding Attacks	5	5	100%
Total	**30**	**30**	**100%**

Results & Evaluation

Overall Performance Summary

Metric	Result
Model Accuracy	93.42%
Red-Team Detection Rate	100% (30/30)
Attack Categories Covered	6
Rule Patterns	30+
Response Time	< 500ms
Training Samples	16,918
Validation Samples	662

Real Scenarios We Tried

Conclusion & Future Scope

What We Created and How It Works

The Sentinel AI stops real threats right now but not someday. This is not a research paper or any idea. It is a live defense system that acts as a firewall and secure the LLM from prompt injection. You get warnings before the damage. Actually the results of this model shows up when they matter mostly.

- Our machine learning system hit 93.42% precision after studying 17000 types of cases. The ML model learned each pattern without skipping any type during training.
- The Red-Team Agent identify threats every single time.
- Invisible speed because of our model keeping latency under 500 ms, so user doesn't face any delay during conversation with LLM model.
- We solve 5 vulnerabilities of the OWASP Top 10 Vulnerabilities of LLM models.
- This system always finding its own flaws . It doesn't wait for any warning or damage.

Who Is Protected and the Way It Works

Industry	What Sentinel AI Does
Healthcare AI	Stops Patient Data Extraction Ensures HIPAA Compliance at AI Level
Finance & Banking	Blocks adversarial prompts targeting fraud detection and trading strategies
EdTech	Keeps learners from tricking study bots into spilling test solutions
Customer Support	Bad replies from AI can damage company's reputation and loose the trust of users. So Taking control of the system to filter the outputs
Government & Defense	We make sure not anyone can steal data from our records. Sentinel AI improves the security enough to stop data leakage, So sensitive information stays safe

Future Scope

Enhancement	Why It Matters
Transformer upgrade DistilBERT DeBERTa	Better detection of subtle and previously unseen attacks
Input Sanitization Layer	In this layer we sanitize the input if the system warns or block them, and after sanitization LLM generate ethical knowledge regarding the input
Role Based Access Control (RBAC)	Different security rules for different user types
Self-learning defense	System retrains automatically when new attack patterns appear
Multi-LLM support	Compatible with GPT Gemini Claude and more beyond Groq
SIEM Integration	Connects to Splunk and QRadar for security operations
SOC 2 and HIPAA certification	A path unfolds into sectors with strict oversight